

Section 1: Business Context

A telecom company is losing customers at an alarming rate. In data collected from 7,043 customers, the churn rate stands at 26.5%, i.e., 1 in every 4 customers leaves. This analysis identifies the key drivers of churn and recommends targeted retention strategies to reduce churn, protect revenue, and strengthen long-term competitiveness. Without intervention, continued customer loss risks significant revenue erosion and long-term competitive decline.

Section 2: Key Findings

- 1. Customer attrition peaks during the initial year of service, marking the first 12 months as the primary window for proactive retention engagement.
- 2. Shorter-term, month-to-month subscribers exhibit significantly higher departure rates compared to long-term contract holders, highlighting contract migration as a high-impact strategic lever.
- 3. Fiber optic users utilizing electronic check payments show a disproportionate churn risk, likely driven by perceived service issues or transactional friction points.
- 4. A segment of 605 High and Critical risk individuals has been pinpointed; with Critical tier churn reaching 67.8%—nearly 16 times the Low tier rate—this group demands immediate budgetary priority.

Section 3: Model Performance

Metric	Before Optimization	After Optimization
Overall Accuracy	Baseline	Improved
Churners Correctly Identified	51%	81%
Missed Churners (False Negatives)	183	70

Explanation:
1. The model's ability to correctly identify at-risk customers improved from 51% to 81% meaning 8 out of every 10

customers likely to churn are now flagged before they leave.

- 2. Missed churners declined from 183 to 70, allowing us to proactively target 113 more at-risk customers for the retention team, each of which is recoverable monthly revenue.

Section 4: Customer Risk Segmentation

Risk Tier	Customers	Actual Churn Rate
Critical	270	67.8%
High	335	36.1%
Medium	217	20.7%

Low	587	4.3%
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- The Critical tier represents the most urgent retention priority - with a 67.8% churn rate, nearly 7 in every 10 customers in this group are at immediate risk of leaving
- Combined, the **605 high** and **critical-risk customers** represent the highest-value targets for retention spend, where proactive outreach will have the greatest measurable impact on revenue protection.
- By segmenting customers into risk tiers rather than treating the entire base uniformly, the retention team can allocate budget precisely - focusing resources where they matter most rather than spreading them thin across 7,043 customers..

Section 5: Recommendations

1. Prioritize contract upgrades for month-to-month customers in the Critical and High-risk tiers, shifting even a fraction to annual contracts directly reduces the highest-churn customer segment.
2. Implement a structured onboarding program for customers in their first 12 months, proactive check-ins, usage guidance, and early incentives can significantly reduce churn during the highest-risk window.
3. Conduct an immediate review of fiber optic service quality - address reliability and satisfaction issues ,and offer targeted retention offers to high-risk fiber optic customers before they leave.
4. Incentivize migration from electronic check to automatic payment methods - customers on auto-pay show significantly lower churn rates, making this a low-cost, high-impact retention lever.
5. Introduce value bundles targeting high-risk customers with multiple services - personalized offers that increase perceived value reduce the likelihood of cancellation and strengthen long-term loyalty.

Section 6: Business Impact Estimate

Based on conservative assumptions \$65 average monthly revenue per customer, a \$50 retention campaign cost, and a 30% retention success rate - targeting the 605 High and Critical risk customers yields an estimated net value of approximately \$111,000 in protected annual revenue.

Without a targeted approach, the business faces two costly alternatives: doing nothing and absorbing the full churn loss, or running broad retention campaigns across all 7,043 customers at significantly higher cost and lower conversion. Risk-tiered targeting eliminates this inefficiency.

Note: Revenue figures are estimated based on industry benchmarks. Actual impact will vary based on company-specific pricing and campaign execution.